

École Normale Supérieure - Physics Department

Parabolic Flight 2022

CNES - NOVESPACE

Influence of gravity on confined Faraday waves





Poitier Vincent - Bacq Coraline - Bourgeois Thibault - Demouchy Mathis - Kassid Abderrahmane

Thursday October 28, 2021

Contents

1 Our team, the project and the schedule **1**

2 Introduction: The Faraday Instability **2**

3 The main experiment: Impact of gravity on Faraday waves **4**

4 Faraday waves on a sphere (Bonus experiment!) **5**

5 Our technical approach **5**

6 Contact **6**

1 Our team, the project and the schedule

We are a team of five people. All of us major in physics and are on our 3rd year of license at the Ecole Normale Supérieure :

- Bacq Coraline, **20** years old, ROLE: **In charge of the conception of the experimental set**
- Bourgeois Thibault, **20** years old, ROLE: **Administrative Manager**
- Demouchy Mathis, **19** years old, ROLE: **Technical Manager**
- Kassid Abderrahmane, **19** years old, ROLE: **In charge of Progamming**
- Poitier Vincent, **21** years old, ROLE: **Communication Manager**

Our project aims towards the study of the effect of gravity in Faraday instabilities in two different situations (great depth state and lesser depth state) : capillary state or gravitational state. This project will also explore the various patterns emerging from these instabilities, as discussed in the following articles :

- "Patterns and quasi-patterns in the Faraday experiment" by W.S.Edwards and S. Fauve. [3]
- "Parametrically excited quasicrystalline surface waves" by W.S.Edwards and S. Fauve. [2]

We are happily supported by two of our professors from the Physics Department of the Ecole Normale Supérieure of Paris : **Lyderic Bocquet** and **Stephan Fauve**.

We will be able to use the "**Fablab**" of the Ecole Normal Superieure of Paris, which is a laboratory where students and professors meet to build prototypes for experiments. We will then be able to engineer and test the experiments before boarding them into the airplane. Our school let us use the laboratory and its equipment for the whole project whenever we want.
After retrieving and analyzing our data, we plan on doing a lot of different things :

- Write a complete article in the *American Journal of Physics* (AJP)
- Write an article in the Normal Physics Review (NPR) which is a journal written by some of the physicists of our Ecole.
- Share our results in events like "Fête de la Science" in France.
- Popularize our experiments in Schools and High Schools in Paris and its suburbs.
- Talk about our experience at CNES during the project and encourage other students to be part of the experience. We will be able to share our experience with other physicists of our school but also with biologists and engineering students part of PSL University (MINES Paristech).
- Compare our results with the ones obtained in two recent articles : "*Faraday waves on band pattern under zero gravity conditions*" [4] **and** "*Influence of capillarity and gravity on confined Faraday waves*" [1].

We plan to spread the work during the year as follows :

- November (after selection) - December : Gathering and organizing all the equipment needed for the experiment (organizing the **Fablab**)
- January : First tests of our experiment with simple shapes of containers.
- February - March : Seek for any improvements of our experiment - Validation of the physical model when gravity dominates over capillarity.
- April - July : Final tests and adjustments for the integration of the experiment into the airplane.

We follow the same courses, so we will be able to work together most of the time. Particularly, we are available to work on the project every Monday morning and on the Weekends.

2 Introduction: The Faraday Instability

When a liquid is subjected to sinusoidal vibrations, there is a phenomenon that might emerge at the surface level under specific conditions, consisting of stationary waves. This is what we call a Faraday instability, which was discovered by Michael FARADAY who mentioned it in an appendix of an article in the *Philosophical Transactions of the Royal Society of London* in **1831**.

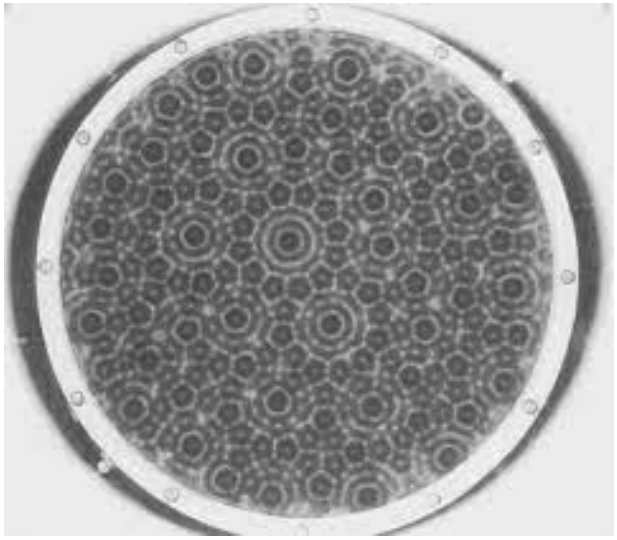


Figure 1: The Faraday Instability

These waves can be characterized theoretically by the following equation :

$$\boxed{\omega^2 = \left(gk + \frac{\gamma k^3}{\rho}\right)\tanh(hk)} \tag{1}$$

where :

- ω is the pulse [rad.s^{-1}]
- g is the gravitational acceleration [m.s^{-2}]
- k is the wave vector [rad.m^{-1}]
- γ is the fluid's surface tension [N.m^{-1}]
- ρ is the fluid's density [kg.m^{-3}]
- h is the container's depth [m]

It appears to us that gravity does impose, alongside capillarity, the specific behavior adopted by the waves. We can rewrite this equation using the capillary length $l_c = \sqrt{\gamma}/\sqrt{\rho g}$ yielding:

$$\omega^2 = \left(l_c^{-2} + k^2\right)\tanh(hk) \tag{2}$$

When the experiment is performed normally in a laboratory on Earth (or any planets with $g \neq 0$!), the gravity term in the equation (2) is more significant than the capillary term. Hence the first goal of our experiment :

Observing the capillary effects, as well as the gravitational effects on Faraday instabilities using the variation of the gravitational acceleration.

We are going to perform this experiment in **two parts** : the *first* part will consist of using a container which depth is smaller than the wavelength λ of the Faraday waves. We then obtain the following equation :

$$\omega^2 \approx hk\left(gk + \frac{\gamma k^3}{\rho}\right)$$

In the *second* part, we will consider using a container with a depth much greater than the wavelength, thus obtaining a different equation :

$$\omega^2 \approx \left(gk + \frac{\gamma k^3}{\rho} \right)$$

The difference between the two configurations can be theoretically obtained by changing the value of l_c, *i.e.* changing the gravitational acceleration.

In the case of the **first approximation**:

- if $l_c << k^{-1}$, using the equation, we notice that k^2 is proportional to ω^2, *i.e.* the relationship between the pulse and the wave vector is linear.
- if $l_c >> k^{-1}$ we find a different relationship : k^4 is proportional to ω^2, *i.e.* there is a power law (square root) linking the pulse to the wave vector.

We get the following FIGURE :

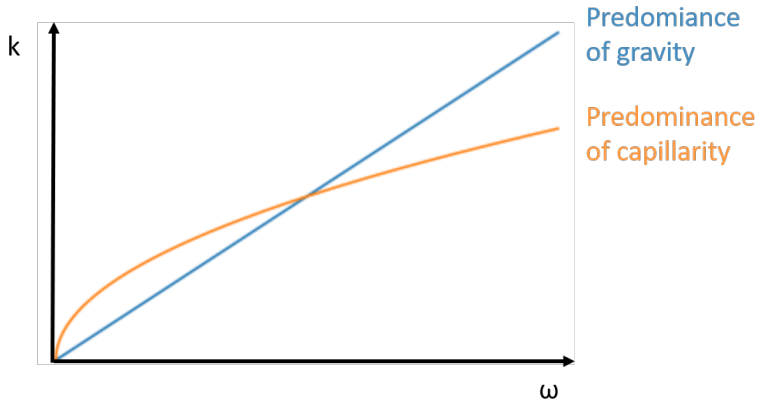


Figure 2: Variations of the wave vector as a function of the pulse (first approximation)

Whereas in the case of the **second approximation**:

- if $l_c << k^{-1}$, using the equation, we notice that k is proportional to ω^2, *i.e.* the relationship between the pulse and the wave vector is a power law (square).
- if $l_c >> k^{-1}$ we find a different relationship : k^3 is proportional to ω^2, *i.e.* there is a power law 2/3 linking the pulse to the wave vector.

We then get :

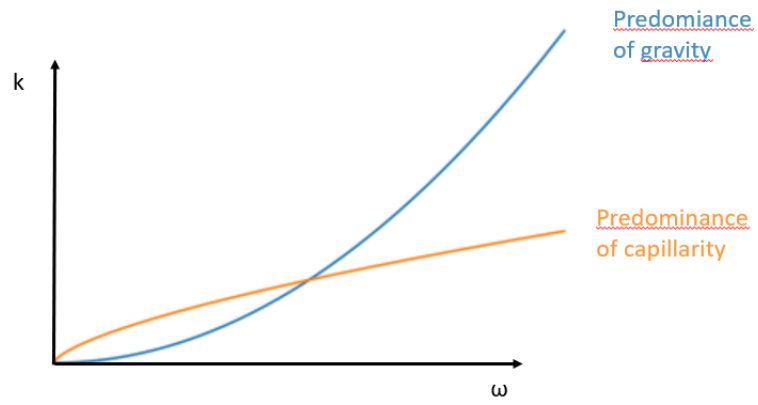


Figure 3: Variations of the wave vector as a function of the pulse (second approximation)

3 The main experiment: Impact of gravity on Faraday waves

Our goal is to investigate the influence of changing the gravitational acceleration on the wave length and shape of Faraday waves. To do that, we will place an container on a vibrating plate. A camera placed on top of the experiment will allow us to retrieve data from the experiment.

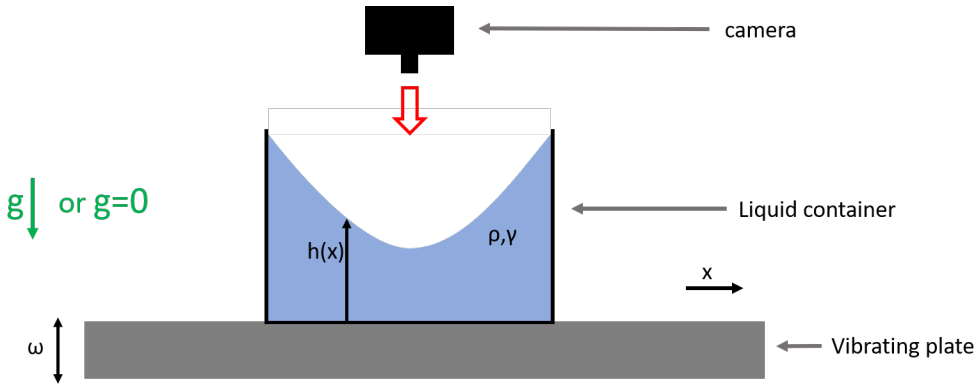


Figure 4: Diagram of the experiment

During each parabola, we will apply to the system an increasing pulse with the vibrating plate to explore various Faraday wavelengths. (We will do that two times, during the hypergravity phase and during the microgravity phase to acquire different values of g).

Moreover, since the wavelength varies depending on the the depth of the container and because in microgravity, the surface of the fluid will not be flat, this wavelength will change as a function of the position in the container. In order to discard this effect, we could build a container with the same shape as the free surface of the fluid. It has also been shown that the shape of the container affects the *eigenmodes* of the Faraday waves. Therefore, we could use different containers in our experiment.

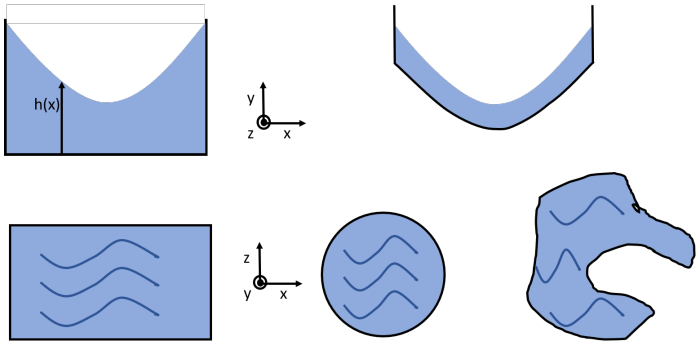


Figure 5: Example of containers that will be used during the flight

4 Faraday waves on a sphere (Bonus experiment!)

Thanks to the lack of gravity, we can manage to isolate a fluid with a spherical shape. By putting said liquid in a container coated with a hydrophobic material except for the vibrator, the fluid with eventually take the shape of a sphere without any disturbance.

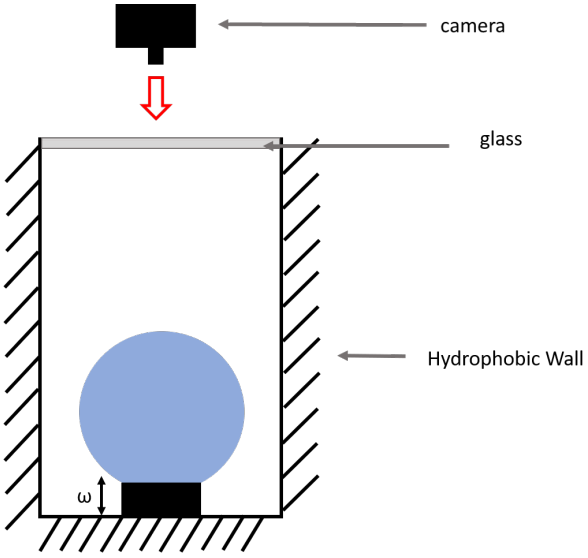


Figure 6: Design of the second experiment

Using this device, it will be possible for us then to study the shift in the Faraday waves' behavior using a different experimental set compared to the previous ones. Moreover, the fact that the liquid's surface is not bound to another material will allow the capillary waves that are surrounding it to interfere, thus changing the eigenmodes previously observed.

5 Our technical approach

Given the restrictions imposed by the CNES, we have come up with a convenient estimate of the equipment that is going to be needed in our experiment :

- Vertical shaker of an estimated mass of 25 Kg (15 cm diameter, 30 cm high)
- Red Pitaya kit
- Four containers of different shapes made of plexiglas (10 cm deep).
- One container coated with a hydrophobic material (such as teflon)
- Fluid (either water or Siliconol, 1 liter)
- Small camera

- Strobe light bulb
- One computer

The electrical consumption of our experimental set is estimated to be 220 V / 4 A.

6 Contact

We are a team of **five** students at Ecole Normale Superieure and we will be more than happy to answer any of your questions about our project :

Poitier Vincent / vincent.poitier@ens.psl.eu
Bacq Coraline / coraline.bacq@ens.psl.eu
Bourgeois Thibault / thibault.bourgeois@ens.psl.eu
Demouchy Mathis / mathis.demouchy@ens.psl.eu
Kassid Abderrahmane / abderrahmane.kassid@ens.psl.eu

References

[1] S. V. Diwakar et al. "Influence of capillarity and gravity on confined Faraday waves". In: *Phys. Rev. Fluids* 3 (7 July 2018), p. 073902. DOI: 10.1103/PhysRevFluids.3.073902. URL: https://link.aps.org/doi/10.1103/PhysRevFluids.3.073902.

[2] S. Fauve W. S. Edwards. "Parametrically excited quasicrystalline surface waves". In: *Physical Review journals* 47.2 (1993). DOI: https://doi.org/10.1103/PhysRevE.47.R788.

[3] S. Fauve W. S. Edwards. "Patterns and quasi-patterns in the Faraday experiment". In: *Journal of Fluid Mechanics* 278 (1994), pp. 123–148. DOI: https://doi.org/10.1017/S0022112094003642.

[4] A. Ivantsov T. Lyubimiva. "Faraday waves on band pattern under zero gravity conditions". In: *Physical Review Fluids, American Physical Society* (2019). DOI: 10.1103/PhysRevFluids.4.064001.

ENS | PSL

Département de Physique
École Normale Supérieure

cnes
CENTRE NATIONAL D'ETUDES SPATIALES